E-commerce Warehouse Location Optimization

A Mixed-Integer Linear Programming Approach

Collaborative academic project.

1 Problem

An e-commerce operator must decide where to place a small number of warehouses to serve its highest-demand cities at minimum cost. Each candidate location has a known annual setup cost and a known daily order volume, and shipping from a warehouse to a city incurs a per-order cost that grows with distance. Opening more warehouses lowers delivery cost but raises setup cost, so the two must be traded off against each other under practical limits on budget, warehouse count, and delivery time.

Concretely: across India's top 10 e-commerce cities, choose at most $K = 4$ warehouse locations and assign every city to an open warehouse so as to minimize total annual cost, subject to an annual setup budget of INR 10,000,000 and a maximum delivery time of 36 hours.

2 Optimization Model

We model this as a *capacitated-free facility location problem* and solve it as a mixed-integer linear program (MILP).

Sets. I: candidate warehouse locations. J: demand cities. Here $I = J$ (the same 10 cities).

Parameters.

- c_i: annual setup cost of a warehouse at location i.
- d_{ij}: per-order delivery cost from i to j, equal to $0.1 \times \text{dist}(i, j)$ (INR per order).
- q_j: expected daily order volume at city j (deterministic).
- t_{ij}: travel time from i to j, equal to $\text{dist}(i, j)/40$ hours.
- $B = 10{,}000{,}000$; $K = 4$; $T = 36$ hours; $D = 365$ (days per year).

Decision variables.

$$x_i \in \{0, 1\} : \text{warehouse opened at } i, \qquad y_{ij} \in \{0, 1\} : \text{city } j \text{ served by } i.$$

Objective. Minimize the total *annual* cost, with daily delivery annualized so that it is on the same time basis as the annual setup cost:

$$\min \underbrace{\sum_{i \in I} c_i \, x_i}_{\text{annual setup}} + \underbrace{D \sum_{j \in J} \sum_{i \in I} d_{ij} \, q_j \, y_{ij}}_{\text{annual delivery}} .$$

Constraints.

$$\sum_{i \in I} c_i \, x_i \le B \qquad \text{(setup budget)} \tag{1}$$

$$\sum_{i \in I} x_i \le K \qquad \text{(warehouse cap)} \tag{2}$$

$$\sum_{i \in I} y_{ij} = 1 \qquad \forall j \in J \quad \text{(each city served once)} \tag{3}$$

$$y_{ij} \le x_i \qquad \forall i, j \quad \text{(serve only from open warehouse)} \tag{4}$$

$$y_{ij} = 0 \qquad \text{if } t_{ij} > T \quad \text{(delivery-time limit)} \tag{5}$$

The key correction over an earlier version of this model is that setup cost appears in the *objective*, not only in the budget constraint. Minimizing delivery cost alone always favors opening the maximum number of warehouses; including setup cost makes the number and choice of warehouses a genuine economic tradeoff. Both terms are expressed on an annual basis so that the reported total is a single, consistent quantity.

3 Data

Distances are computed with the Haversine (great-circle) formula on city coordinates; delivery cost and time are derived from distance using the rate 0.1 INR /km/order and a speed of 40 km/h. Demand is taken as the expected daily order volume per city.

Some of the input values (including certain setup costs, demand figures, and rate assumptions) are illustrative placeholders rather than sourced data, chosen to exercise the model. The purpose of this study is to demonstrate that the formulation and solver produce a sensible, constraint-satisfying solution; the absolute cost figures should be read as model outputs on these inputs, not as empirical estimates.

City	Demand (orders/day)	Setup cost (INR)	Coordinates (lat, lon)
Mumbai	1500	3,000,000	(19.076, 72.878)
Delhi	1200	2,500,000	(28.704, 77.103)
Bangalore	1000	2,000,000	(12.972, 77.595)
Hyderabad	800	1,800,000	(17.385, 78.487)
Ahmedabad	600	1,500,000	(23.023, 72.571)
Chennai	700	2,200,000	(13.083, 80.271)
Kolkata	900	2,400,000	(22.573, 88.364)
Pune	500	1,700,000	(18.520, 73.857)
Jaipur	300	1,200,000	(26.912, 75.787)
Lucknow	200	1,300,000	(26.847, 80.946)

Table 1: City parameters.

4 Method

The MILP is formulated and solved in Python using `PuLP` with the CBC solver. For each ordered pair of cities the Haversine distance is computed, from which per-order delivery cost and travel time follow. For each city, only warehouses reachable within the time limit T are considered as feasible sources. The solver returns the set of open warehouses and the city-to-warehouse assignment that minimize total annual cost.

5 Results

The solver returns an optimal solution opening four warehouses.

Metric	**Value**
Selected warehouses	Mumbai, Delhi, Bangalore, Kolkata
Number of warehouses	4
Daily delivery cost	INR 1,08,461
Annual delivery cost	INR 3,95,88,147
Annual setup cost	INR 99,00,000
Total annual cost	**INR 4,94,88,147**
Solver status	Optimal

Table 2: Optimal solution (annual basis).

City	**Served by**	**Cost/order (INR)**	**Delivery time (hrs)**
Mumbai	Mumbai	0.00	0.00
Delhi	Delhi	0.00	0.00
Bangalore	Bangalore	0.00	0.00
Kolkata	Kolkata	0.00	0.00
Pune	Mumbai	12.02	3.00
Jaipur	Delhi	23.75	5.94
Chennai	Bangalore	29.02	7.25
Lucknow	Delhi	43.08	10.77
Ahmedabad	Mumbai	44.00	11.00
Hyderabad	Bangalore	50.00	12.50

Table 3: City assignments. Self-served cities incur no inter-city delivery cost or time under this model.

The four selected cities act as regional hubs: Mumbai covers the west (Pune, Ahmedabad), Delhi the north (Jaipur, Lucknow), Bangalore the south (Chennai, Hyderabad), and Kolkata the east. Kolkata is selected despite its moderate demand because it is geographically isolated from the other hubs; serving it from any other warehouse would incur a large delivery cost.

6 Discussion and Limitations

The result rests on several simplifying assumptions, stated here so the model's scope is clear:

- **Intra-city delivery is free and instant.** A self-served city has zero delivery cost and time. In practice intra-city fulfilment has a nonzero cost; adding a fixed intra-city term would raise the reported figures but is unlikely to change the selection.
- **Great-circle distance.** Haversine underestimates true road distance. A detour factor (roughly 1.4 for Indian road networks) would scale all delivery costs and times upward. The delivery-time constraint is slack here (the longest assignment is 12.5 hours against a 36-hour limit), so this does not affect feasibility, but it does mean the cost figures are optimistic.
- **Single sourcing.** Each city is served by exactly one warehouse. Allowing demand splitting could lower cost but complicates the model.
- **Deterministic demand.** Demand is the expected daily order volume, held fixed. Day-to-day variability (for example festival spikes) is not modeled; a stochastic formulation is

left to future work.

- **No warehouse capacity limit.** A warehouse can serve unlimited demand. Adding a capacity constraint would make the problem a fully capacitated facility location problem.

7 Conclusion

Formulating warehouse placement as a MILP and minimizing a single, consistently defined annual total cost, the model selects Mumbai, Delhi, Bangalore, and Kolkata as warehouse locations, covering all ten demand cities within the budget, warehouse-count, and delivery-time constraints at a total annual cost of approximately INR 4.95 crore on the given inputs. The main value of the exercise is the clean separation of the setup-versus-delivery tradeoff; the limitations above, together with the placeholder nature of some inputs, indicate the natural directions for a more realistic model.